ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

BIOLOGY

PAPER 1 Multiple Choice

4025/1

NOVEMBER 2023 SESSION

1 hour

Additional materials:
- Multiple Choice answer sheet
- Soft clean eraser
- Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer.

Any rough working should be done in this booklet.

This question paper consists of 16 printed pages.

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1. Which triangle colour code represents a poisonous substance?

A orange

B green

C purple

D red

2. The application of biological systems in industry is called

A microbiology.

B biotechnology.

C anatomy.

D physiology.

3. Water has high specific heat capacity.
This is important for

A photosynthesis.

B translocation.

C cooling.

D transpiration.

4. Which result indicates positive for the presence of lipids in a food test?

A blue black

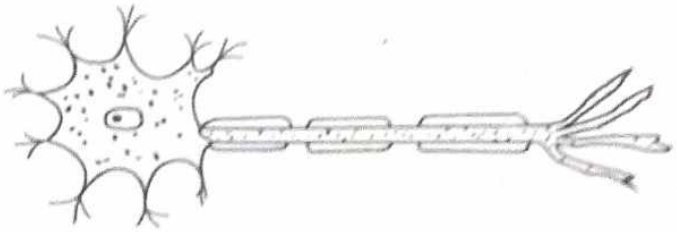
B brick red

C purple mauve

D cloudy white

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5. The diagram shows a specialised cell.



The cell shown is a

A root hair cell.

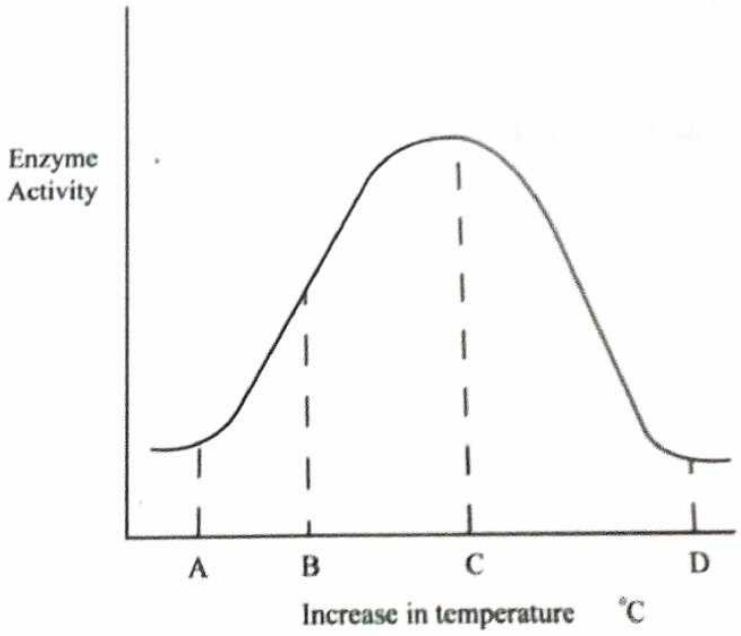
B white blood cell.

C red blood cell.

D nerve cell.

6. The graph shows the effect of temperature on the activity of a mammalian enzyme action.

At which point, **A**, **B**, **C** or **D**, would the enzyme be denatured?





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7. Enzymes are affected by pH because they are

A biological catalysts.

B specific in nature.

C protein in nature.

D coded for by DNA.

8. Which one is a function of nitrogen in plants?

A root development

B flower formation

C ATP synthesis

D protein synthesis

9. A carnivorous beetle can be used to control aphids in a garden. Which method of pest control is this?

A biological control

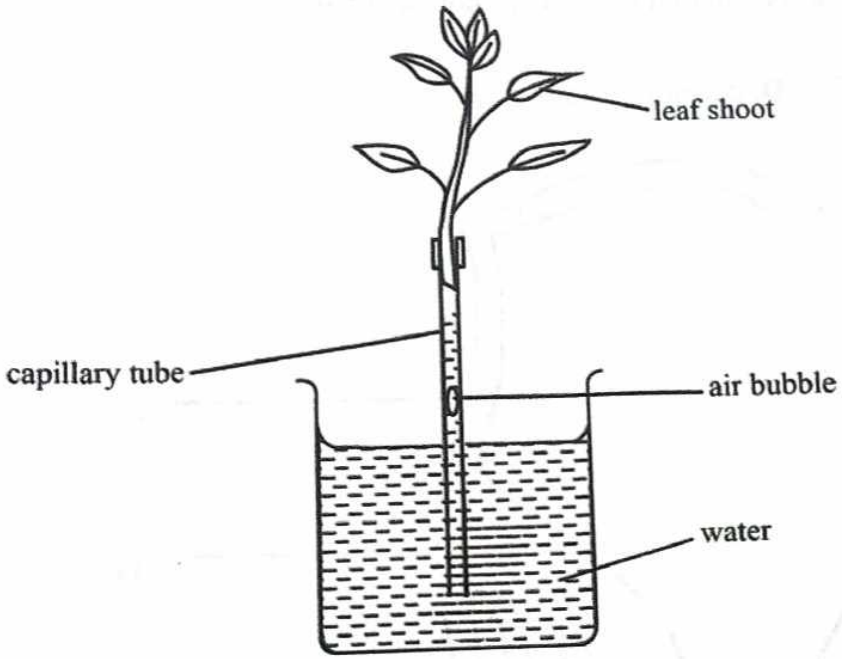
B chemical control

C cultural control

D mechanical control

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10. The diagram shows a simple potometer to measure water uptake.



When placed in windy conditions, the bubble

A moves downwards.

B remains stationary.

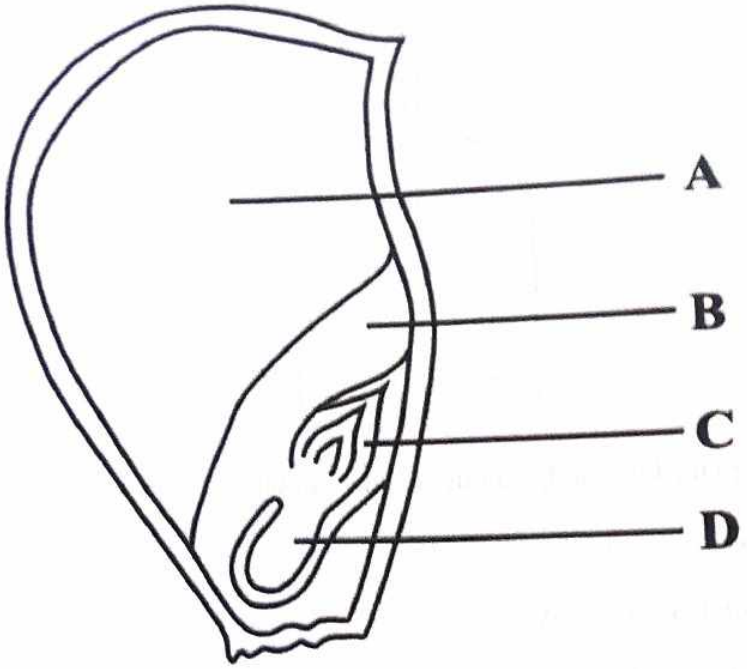
C increases in size.

D moves upwards.

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11. The diagram shows a section of a monocotyledonous seed.

Which part, **A**, **B**, **C** or **D**, develops into a root?



12. Which growth hormone is released by plants in response to gravity?

A ethylene

B cytokinin

C gibberellin

D auxin

13. Diabetes mellitus is a result of the malfunction of the

A pancreas.

B liver.

C gall bladder.

D kidney.

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14. A vaccine contains an

A antigen.

B antibody.

C antibiotic.

D antiretroviral.

15. Which component of blood is responsible for blood clotting?

A serum

B plasma

C platelets

D lymphocytes

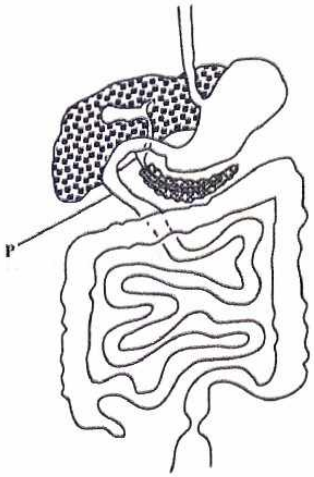
16. A human eye was subjected to different light intensity at **R** and at **S**. At **R**, the circular muscle of the iris contracts and the radial muscle relaxes. At **S** the circular muscle of th iris relaxes and the radial muscle contracts.

Which one is correct about the light intensity at **R** and at **S** respectively?

	R	**S**
A	bright	bright
B	bright	dim
C	dim	bright
D	dim	dim

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17. The diagram shows the human alimentary canal.



The part labelled **P** is the

A rectum.

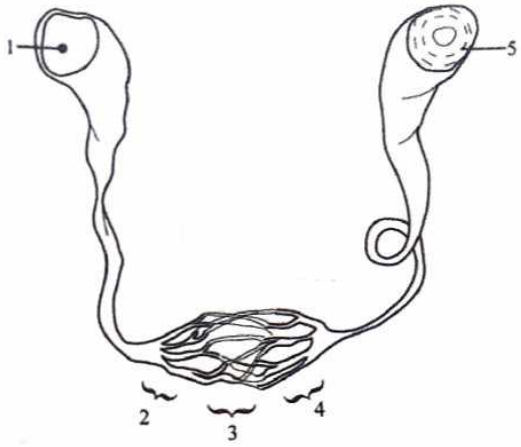
B eosophagus.

C ileum.

D duodenum

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19. The diagram shows blood vessels in an animal body.



Which row **A**, **B**, **C** and **D** correctly identifies the vessels 1, 3 and 5?

	1	3	5
A	artery	capillaries	vein
B	vein	capillaries	artery
C	artery	vein	capillaries
D	capillaries	vein	artery

20. The function of amniotic fluid is

A exchange of materials.

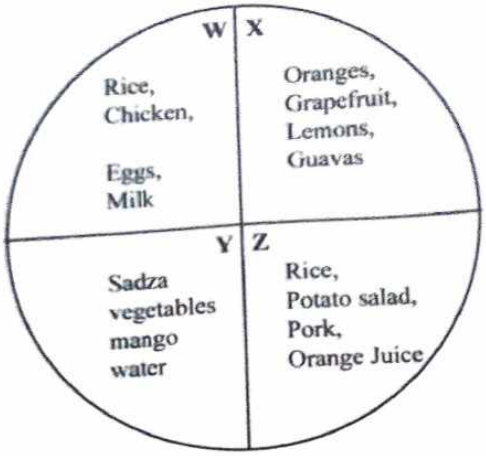
B destroying bacteria.

C maintaining temperature.

D transport hormones.

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21. The diagram shows meals, **W**, **X**, **Y** and **Z**.



Which one is recommended for a mine worker?

A W.

B X.

C Y.

D Z.

22. Insulin stimulates the liver to convert

A fat to glycogen.

B glucose to fat.

C glycogen to glucose.

D glucose to glycogen.

23. Mechanical digestion is

A movement of food along the gut by peristalsis.

B ingestion of food for assimilation by the body.

C breaking large pieces of food into smaller pieces.

D breaking large insoluble molecules into small soluble molecules.

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24. The function of the Eustachian tube found in the ear is to

A maintain balance in an upright position.

B amplify vibrations in the ear canal.

C transmit vibrations to the oval window.

D prevent the damage to tympanic membrane.

25. Which row, **A**, **B**, **C** or **D**, shows the action of muscles to straighten the arm?

	extensor muscle	flexor muscle
A	contracts	relaxes
B	relaxes	contracts
C	contracts	contracts
D	relaxes	relaxes

26. During dialysis, blood cells and protein molecules can not escape from the patient's blood because they

A have high pH.

B are large in size.

C are insoluble in water.

D have a lower concentration.

27. A patient with a head injury has difficulties in walking.
Which part of the brain has been damaged?

A cerebellum.

B hypothalamus.

C medulla.

D pituitary.

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28. A unit of inheritance is

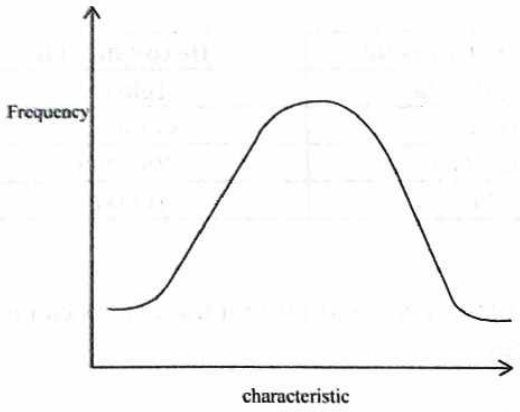
A an allele.

B a gene.

C the DNA.

D the RNA.

29. The graph shows the distribution of a characteristic in a population.



Which characteristic can be represented by the graph?

A complexion

B dimples

C eye colour

D blood group

30. Which condition is a result of a chromosomal mutation?

A albinism

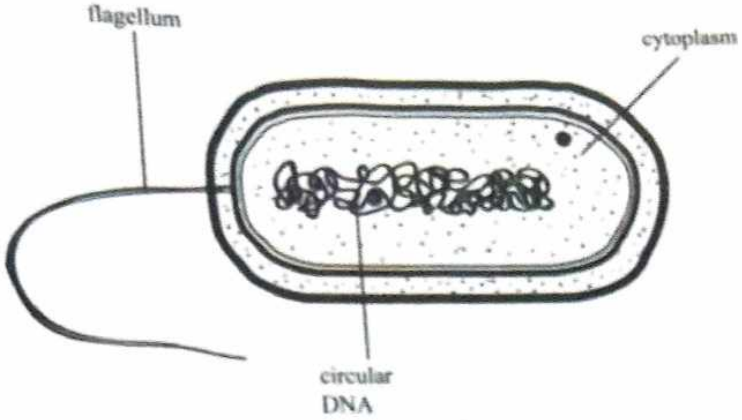
B cancer

C Down's syndrome

D sickle cell anaemia

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31. The diagram shows a microorganism, **Z**.



To which kingdom does **Z** belong?

A Animalia

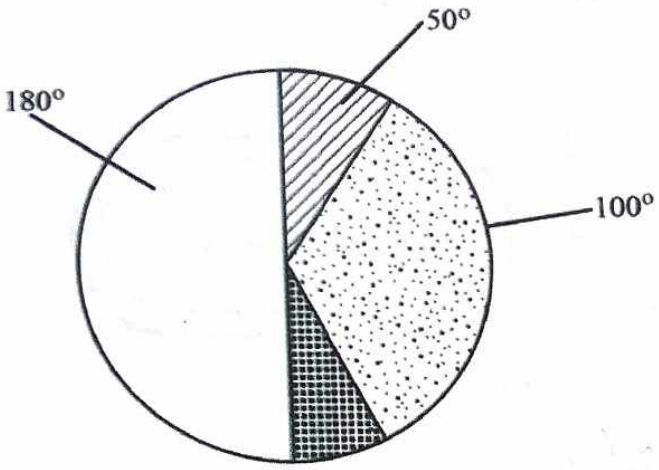
B Prokaryotae

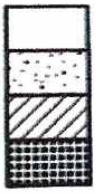
C Fungi

D Protista

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32. The pie-chart shows components of a soil sample.





water
air content
mineral particles
organic matter

The proportion of air in the soil sample is

A 18 %

B 28 %

C 36 %

D 56 %

33. Which one is an environmentally friendly method of waste disposal?

A combustion

B landfill

C recycling

D gasification

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34. The process responsible for the decrease in biomass in the pyramid of biomass is

A photosynthesis.

B respiration.

C decomposition.

D nitrification.

35. The food chain shows energy transfer in a river.

algae → tadpoles → fish → eagles

Which one is an effect of overfishing?

A tadpoles increase, eagles decrease

B tadpoles increase, eagles increase

C tadpoles decrease, algae increase

D tadpoles decrease, algae decrease

36. An ecosystem is

A interdependent living organisms and their physical environment.

B the place living organisms in the physical environment stay .

C different species of living organisms in the physical environment.

D roles played by living organisms in the physical environment.

37. In an ecosystem, energy is transferred from one trophic level to another by

A excretion.

B egestion.

C respiration.

D ingestion.

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38. Which one is an effect of alcohol on a driver?

A increases reaction time

B reduces breaking distance

C increases muscular coordination

D reduces blurred vision

39. Which one is a non-infectious disease?

A cholera

B kwashiorkor

C typhoid

D tuberculosis

40. Which method does **not** prevent the spread of the corona virus?

A isolation

B quarantine

C sanitising

D handshaking

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